



MT9M113: 1/6-Inch 1.3Mp SOC Digital Image Sensor Register Description

Table 32: Driver ID = 11: Histogram Variables (continued)

R/W (Read or Write) bit; RO (Read Only) bit

Reg. #	Bits	Default	Name
R0x0053	15:0	0x0050	hg_gamma_table_b_4 (R/W)
			Gamma table B entry for low light condition
R0x0054	15:0	0x006A	hg_gamma_table_b_5 (R/W)
			Gamma table B entry for low light condition
R0x0055	15:0	0x0080	hg_gamma_table_b_6 (R/W)
			Gamma table B entry for low light condition
R0x0056	15:0	0x0091	hg_gamma_table_b_7 (R/W)
			Gamma table B entry for low light condition
R0x0057	15:0	0x00A1	hg_gamma_table_b_8 (R/W)
			Gamma table B entry for low light condition
R0x0058	15:0	0x00AF	hg_gamma_table_b_9 (R/W)
			Gamma table B entry for low light condition
R0x0059	15:0	0x00BB	hg_gamma_table_b_10 (R/W)
			Gamma table B entry for low light condition
R0x005A	15:0	0x00C6	hg_gamma_table_b_11 (R/W)
			Gamma table B entry for low light condition
R0x005B	15:0	0x00D0	hg_gamma_table_b_12 (R/W)
			Gamma table B entry for low light condition
R0x005C	15:0	0x00D9	hg_gamma_table_b_13 (R/W)
			Gamma table B entry for low light condition
R0x005D	15:0	0x00E2	hg_gamma_table_b_14 (R/W)
			Gamma table B entry for low light condition
R0x005E	15:0	0x00EA	hg_gamma_table_b_15 (R/W)
			Gamma table B entry for low light condition
R0x005F	15:0	0x00F1	hg_gamma_table_b_16 (R/W)
			Gamma table B entry for low light condition
R0x0060	15:0	0x00F9	hg_gamma_table_b_17 (R/W)
			Gamma table B entry for low light condition
R0x0061	15:0	0x00FF	hg_gamma_table_b_18 (R/W)
			Gamma table B entry for low light condition

Table 33: GPIO Registers

R/W (Read or Write) bit; RO (Read Only) bit

Reg. #	Bits	Default	Name
R0x1070	15:0	0x0000	gpio_data (R/W)
	15:13	X	Reserved
	12:8	0x0000	gpio_4_0_data Shows the state of GPIO signals and controls some of those signals when they are configured as outputs. Bits 12 to 8 show/control the state of the signals GPIO[4] to GPIO[0], respectively. If certain GPIO signals are floating during STANDBY, the corresponding bits should be tied to "0" converting the signals to output to reduce standby current.
	7:0	X	Reserved
			Shows the state of the GPIO signals and controls those configured as outputs. The state of all GPIO signals are sampled at power-up (see also gpi_status at R0x0024).



MT9M113: 1/6-Inch 1.3Mp SOC Digital Image Sensor Register Description

Table 33: GPIO Registers (continued)

R/W (Read or Write) bit; RO (Read Only) bit

Reg. #	Bits	Default	Name
R0x1074	15:0	0x0000	gpio_output_set (R/W)
	15:13	X	Reserved
	12:8	0x0000	gpio_4_0_output_set Output goes high when the GPIO is configured as an output. All the bits are read only, and the bits are reset back to "0" after they are written. 1: Corresponding GPIO signal becomes a "1".
	7:0	X	Reserved
	This register sets the desired output GPIO signals high. Only GPIO[3] and GPIO[2] can be controlled using this register.		
R0x1076	15:0	0x0000	gpio_output_clear (R/W)
	15:13	X	Reserved
	12:8	0x0000	gpio_4_0_output_clear Output goes low when the GPIO is configured as an output. All the bits are read only, and the bits are reset back to "0" after they are written. 1: Corresponding GPIO signal becomes a "0".
	7:0	X	Reserved
	This register sets the desired output GPIO signals low. Only GPIO[3] and GPIO[2] can be controlled using this register.		
R0x1078	15:0	0x1F00	gpio_dir (R/W)
	15:13	X	Reserved
	12	0x0001	gpio_4_dir When configured as a flash output, this signal is controlled by the flash configuration variables in the sequencer variables. 0: Flash output 1: General-purpose input
	11	0x0001	gpio_3_dir GPIO[3] can be set as input or output. 0: General-purpose output 1: General-purpose input
	10	0x0001	gpio_2_dir To configure GPIO[2] as OE_BAR, oe_n_enable (R0x001A[8]) needs to be set to "1." 0: General-purpose output 1: OE_BAR
	9	0x0001	gpio_1_dir To display 10-bit output, this bit needs to be set to output as DOUT_LSB1. 0: DOUT_LSB1 1: General-purpose input
	8	0x0001	gpio_0_dir To display 10-bit output, this bit needs to be set to output as DOUT_LSB0. 0: DOUT_LSB0 1: General-purpose input
	7:0	X	Reserved
	Controls the direction of five GPIO signals. Each GPIO can be configured independently to be an input or output. Setting any bit to "0" configures the corresponding GPIO signal as an output, and setting any bit to "1" configures the corresponding GPIO signal as an input.		



MT9M113: 1/6-Inch 1.3Mp SOC Digital Image Sensor Timing Specifications

Timing Specifications

Power-up sequence

Powering up the sensor requires voltages to be applied in a particular order as seen in Figure 29. The timing requirements are shown in Table 34. The sensor includes a power-on reset feature that initiates a reset upon power up, as shown in Figure 30 on page 104. Even though this feature is included on the device, it is advised that the user still manually assert a hard reset upon power up.

Figure 29: Power Up Sequence

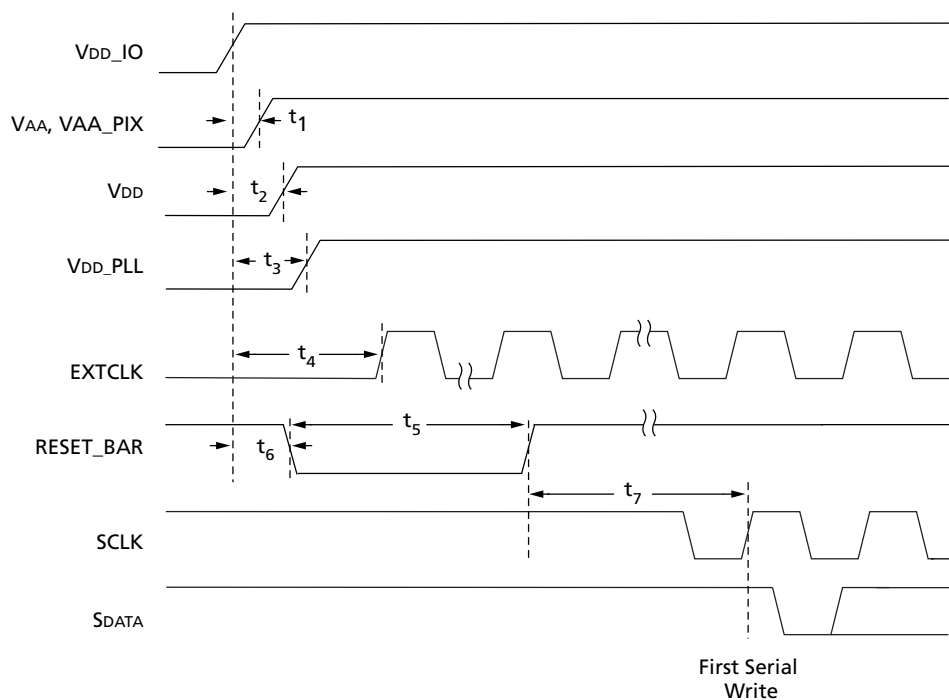


Table 34: Power-Up Signal Timing

Parameter	Symbol	Condition	Min	Typ	Max	Units
Delay from VDD_IO to VAA and VAA_PIX	t_1		0	–	500	ms
Delay from VDD_IO to VDD	t_2		0	–	500	ms
Delay from VDD_IO to VDD_PLL	t_3		0	–	500	ms
EXTCLK Activation	t_4		–	500	–	ms
RESET_BAR initialization time	t_6		–	2	–	ms
RESET_BAR activation time	t_5		–	70	1	EXTCLKs
First serial write	t_7		–	100		EXTCLKs

- Notes:
1. VDD_PLL and VDD_IO should be tied together.
 2. VDD and VDDIO_TX should be tied together.



MT9M113: 1/6-Inch 1.3Mp SOC Digital Image Sensor Timing Specifications

Power on Reset

Figure 30: Power On Reset Sequence

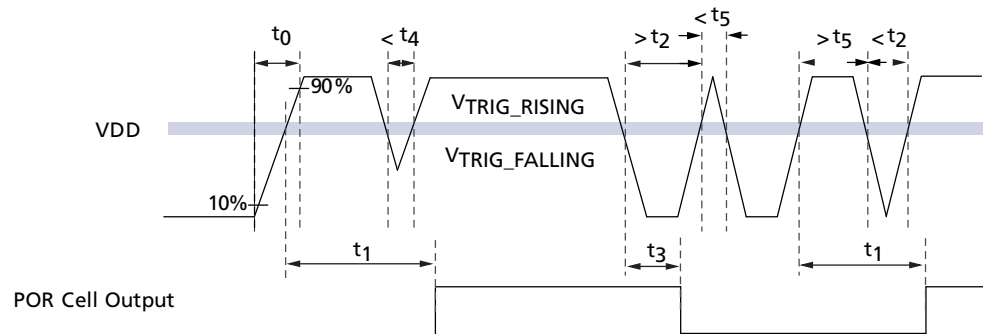


Table 35: POR Parameters

Parameter	Symbol	Condition	Min	Typ	Max	Units
VDD ramp time	t_0		–	–	–	μs
VDD rising, crossing VTRIG_RISING internal reset being released	t_1		7	10	15	μs
VDD falling, crossing VTRIG_FALLING; internal reset active	t_2		–	0.5	1	μs
Minimum VDD spike width below VTRIG_FALLING; considered to be a reset when POR cell output is HIGH	t_3		–	0.5	–	μs
Minimum VDD spike width below VTRIG_FALLING considered to be a reset when POR cell output is LOW	t_4		–	1	–	μs
Minimum VDD spike width above VTRIG_RISING considered to be a stable supply when POR cell output is LOW	t_5		–	50	–	ns
VDD rising trigger voltage	VTRIG_RISING		1.15	1.4	1.55	V
VDD falling trigger voltage	VTRIG_FALLING		1	1.25	1.45	V

Note: The hard reset and POR signals are gated, therefore, M3 ROM READ will begin after both have been de-asserted.

The VDD ramp time required are MAX = 10 μs . If external VDD ramp is more than 10 μs , an external reset is required.



MT9M113: 1/6-Inch 1.3Mp SOC Digital Image Sensor Timing Specifications

Reset

Two types of reset are available:

- A hard reset is issued by toggling RESET_BAR.
- A soft reset is issued by writing commands through the two-wire serial interface.

Hard Reset

After hard reset, the output FIFO is configured for operation but disabled and all outputs are tri-stated. These outputs can be enabled through the two-wire serial interface. The hard reset signal sequence is shown in Figure 31. Hard reset timing is shown in Table 36.

Figure 31: Hard Reset Signal Sequence

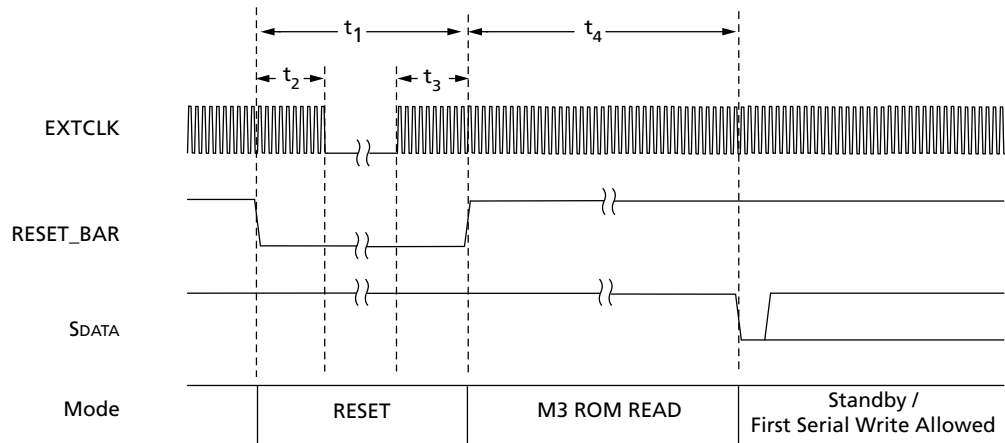


Table 36: Hard Reset Signal Timing

Parameter	Symbol	Condition	Min	Typ	Max	Units
RESET_BAR pulse width	t_1		70	–	–	EXTCLKs
Active ECXTCLK after RESET_BAR asserted	t_2		10	–	–	EXTCLKs
Active EXTCLK before RESET_BAR de-asserted	t_3		10	–	–	EXTCLKs
Max ROM read time	t_4		–	–	6000	EXTCLKs



MT9M113: 1/6-Inch 1.3Mp SOC Digital Image Sensor Timing Specifications

Soft Reset

A soft reset sequence to the sensor has the same affect as the hard reset and can be activated writing to a register through the two-wire serial interface. The soft reset signal sequence is shown in Figure 32. Soft reset timing is shown in Table 37.

Figure 32: Soft Reset Signal Sequence

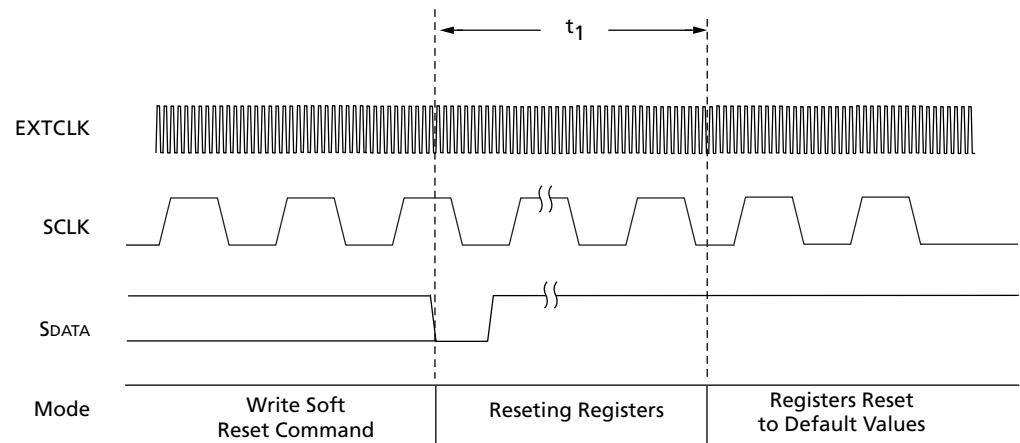


Table 37: Soft Reset Signal Timing

Parameter	Symbol	Condition	Min	Typ	Max	Units
Active EXCLK after soft reset command asserted	t_1		–	6000	–	EXCLKs



MT9M113: 1/6-Inch 1.3Mp SOC Digital Image Sensor Timing Specifications

Standby Modes

The MT9M113 supports two different standby modes:

- Hard standby mode
- Soft standby mode

For each mode, entry can be inhibited by programming the standby_control register.

Hard Standby

Hard standby mode uses STANDBY to shut down digital power (VDD) and ensure the lowest power consumption. All the two-wire serial interface settings and firmware variables including patches will be lost in this mode. Starting up from this mode is equivalent to power up. The two-wire serial interface will be inactive and the sensor must be started up by de-asserting STANDBY. The hard standby signal sequence is shown in Figure 33. Hard standby timing is shown in Table 38.

Figure 33: Hard Standby Signal Sequence

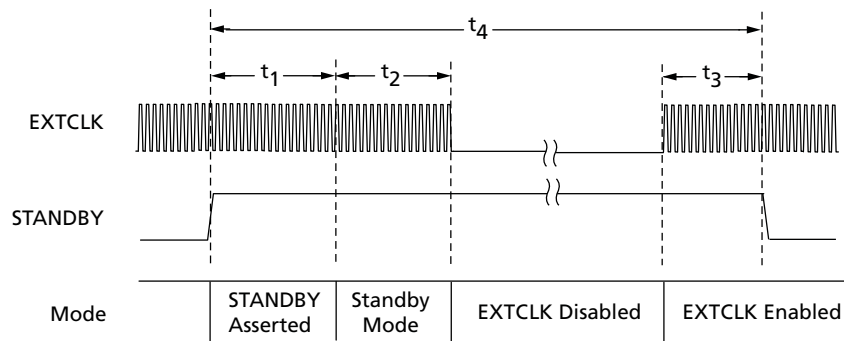


Table 38: Hard Standby Signal Timing

Parameter	Symbol	Condition	Min	Typ	Max	Units
Standby effective delay	t_1		20	–	–	EXTCLKs
Active EXTCLK after STANDBY asserted	t_2		10	–	–	EXTCLKs
Active EXTCLK before STANDBY de-asserted	t_3		10	–	–	EXTCLKs
STANDBY pulse width	t_4		100	–	–	EXTCLKs



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Soft Standby

Soft standby can be enabled by register access and disables the sensor core and most of the digital logic. The two-wire serial interface is still active and the sensor can be programmed through register commands. All register settings and RAM content will be preserved. Soft standby can be performed in any sequencer state after all AE, AWB, histogram, and flicker calculations are finished and the sensor is in low-power mode.

The execution of standby will take place after the completion of the current line by default. It is possible to synchronize the execution of standby with the end of frame through the standby_control register. The soft standby signal sequence is shown in Figure 34. Soft standby timing is shown in Table 39.

Figure 34: Soft Standby Signal Sequence

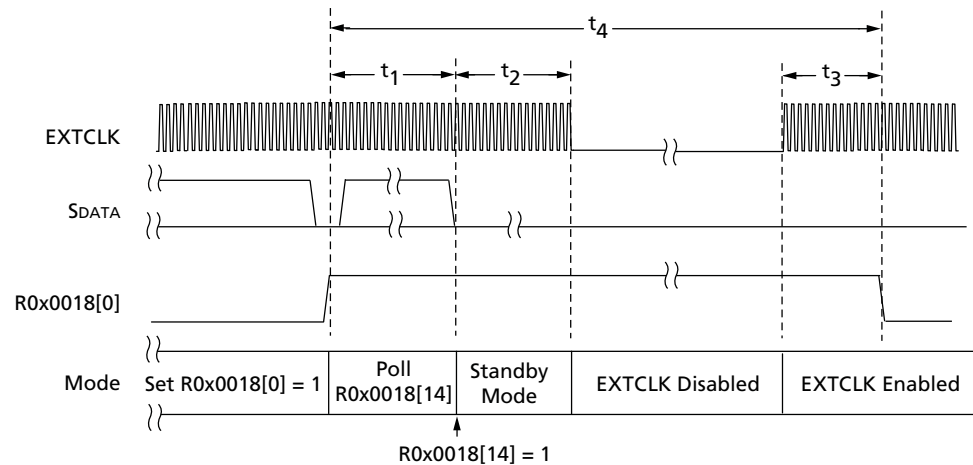


Table 39: Soft Standby Signal Timing

Parameter	Symbol	Condition	Min	Typ	Max	Units
Standby entry complete	t_1		–	20	–	EXTCLKs
Active EXTCLK after soft standby activates	t_2		–	10	–	EXTCLKs
Active EXTCLK before soft standby de-activates	t_3		–	10	–	EXTCLKs
Minimum standby time	t_4		–	100	–	EXTCLKs



MT9M113: 1/6-Inch 1.3Mp SOC Digital Image Sensor Timing Specifications

Signal States

Table 40: Status of Signals During Different States

Signal	Reset	Post-Reset	Standby	Standby with SHUTDOWN	Power Down
DOUT[7:0]	High-Z	High-Z	High-Z by default (configurable via OE_BAR or two-wire serial interface reg)	High-Z by default	X
PIXCLK	High-Z	High-Z	High-Z by default (configurable)	High-Z by default	X
LV	High-Z	High-Z	High-Z by default (configurable)	High-Z by default	X
FV	High-Z	High-Z	High-Z by default (configurable)	High-Z by default	X
DOUT_N	0	0	0	0	X
DOUT_P	0	0	0	0	X
CLK_N	0	0	0	0	X
CLK_P	0	0	0	0	X
GPIO[4:0]	High-Z	High-Z	Depending on how system uses them as DOUT_LSB1/DOUT_LSB2/SHUTTER/FLASH/MODULE_ID/TRIGGER/OE_BAR	Depending on how system uses them as DOUT_LSB/SHUTTER/FLASH/MODULE_ID/TRIGGER/OE_BAR	X
SADDR	Input	Input	Input	Input	
SDATA	Input	I/O	Input	Input	
SCLK	Input	Input	Input	Input	

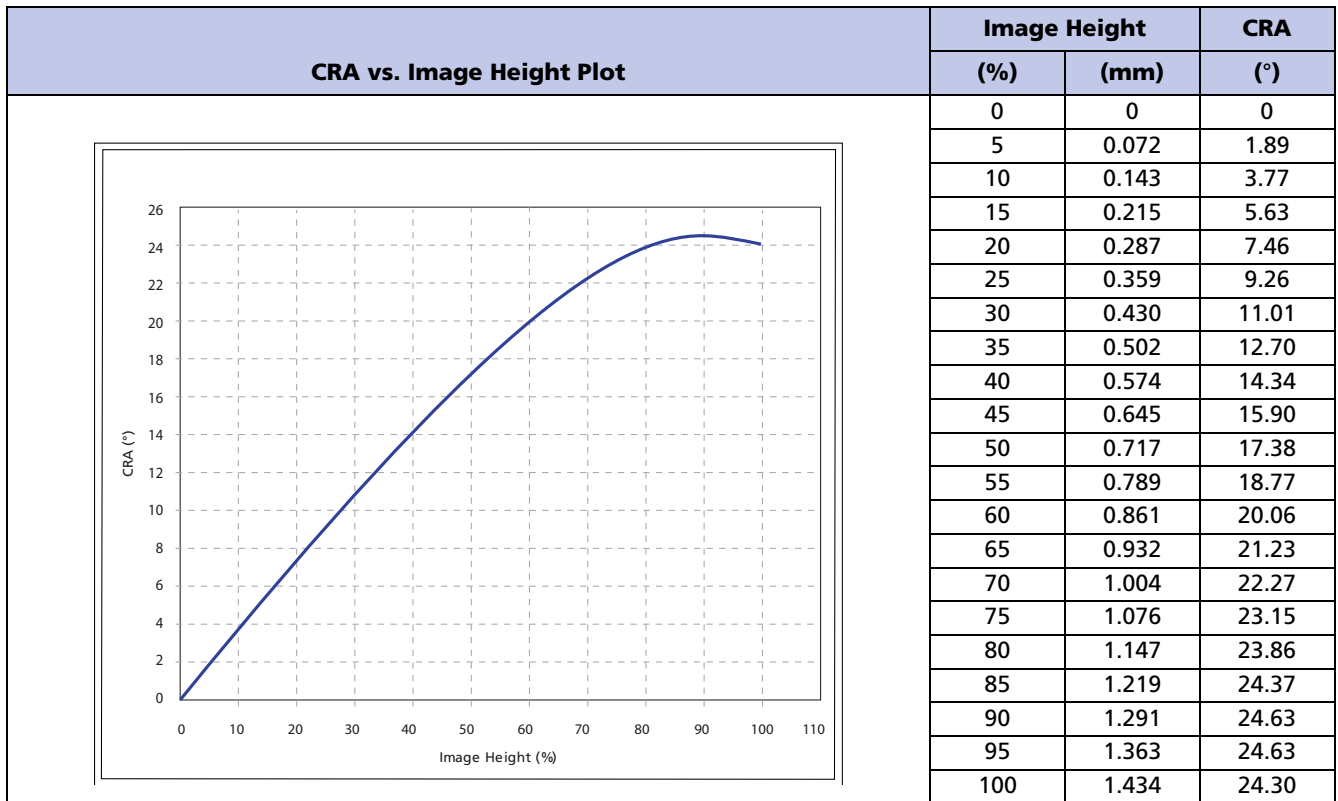
Note: X = "Don't Care."



MT9M113: 1/6-Inch 1.3Mp SOC Digital Image Sensor Spectral Characteristics

Spectral Characteristics

Figure 35: Chief Ray Angle (CRA) vs. Image Height

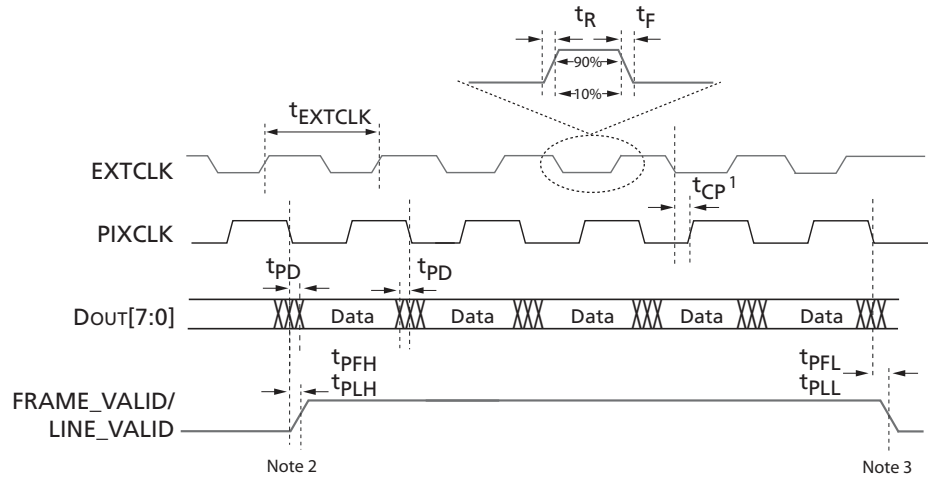




MT9M113: 1/6-Inch 1.3Mp SOC Digital Image Sensor Electrical Specifications

Electrical Specifications

Figure 36: I/O Timing Diagram



- Notes:
1. PLL disabled for t_{CP} .
 2. FV lead LV by 6 PIXCLKs.
 3. FV trails LV by 6 PIXCLKs.

Table 41: I/O Timing Specifications

$f_{EXTCLK} = 8\text{--}54\text{ MHz}$; $V_{DD} = 1.8\text{V}$; $V_{DD_IO} = 1.8\text{V--}2.8\text{V}$; $V_{AA} = 2.8\text{V}$; $V_{AA_PIX} = 2.8\text{V}$;
 $V_{DD_PLL} = 2.8\text{V}$; $V_{DD_PHY} = \text{NA}$; $T_J = 25^\circ\text{C}$; $C_{LOAD} = 15\text{--}20\text{pF}$

Parameter	Symbol	Conditions	Min	Typ	Max	Units	Notes
External clock frequency	f_{EXTCLK}	PLL enabled	8	–	54	MHz	
External clock period	t_{EXTCLK}	PLL enabled	18.5	–	125	ns	
Input clock rise time	t_R		–	2	5	V/ns	
Input clock fall time	t_F		–	2	5	V/ns	
Clock duty cycle			40	50	60	%	
Input clock jitter (peak-peak cycle jitter)	t_{JITTER}		–	0.5	1	ns	1
EXTCLK rising to PIXCLK propagation delay	t_{CP}		5	$0.25 \cdot t_{EXTCLK}$	–	ns	2



MT9M113: 1/6-Inch 1.3Mp SOC Digital Image Sensor Electrical Specifications

Table 41: I/O Timing Specifications (continued)

^fEXTCLK = 8–54 MHz; V_{DD} = 1.8V; V_{DD_IO} = 1.8V–2.8V; V_{AA} = 2.8V; V_{AA_PIX} = 2.8V;
V_{DD_PLL} = 2.8V; V_{DD_PHY} = NA; T_J = 25°C; C_{LOAD} = 15–20pF

Parameter	Symbol	Conditions	Min	Typ	Max	Units	Notes
No programmable slew	Output pin slew	MAX V _{DD_IO} CLOAD = 15pf	TBD	–	TBD	V/ns	
		TYP V _{DD_IO} CLOAD = 30pf	TBD	–	TBD	V/ns	
		MIN V _{DD_IO} CLOAD = 60pf	TBD	–	TBD	V/ns	
Programmable slew condition 1		MAX V _{DD_IO} CLOAD = 15pf	TBD	–	TBD	V/ns	
		TYP V _{DD_IO} CLOAD = 30pf	TBD	–	TBD	V/ns	
		MIN V _{DD_IO} CLOAD = 60pf	TBD	–	TBD	V/ns	
Programmable slew condition 2		MAX V _{DD_IO} CLOAD = 15pf	TBD	–	TBD	V/ns	
		TYP V _{DD_IO} CLOAD = 30pf	TBD	–	TBD	V/ns	
		MIN V _{DD_IO} CLOAD = 60pf	TBD	–	TBD	V/ns	
Programmable slew condition n		MAX V _{DD_IO} CLOAD = 15pf	TBD	–	TBD	V/ns	
		TYP V _{DD_IO} CLOAD = 30pf	TBD	–	TBD	V/ns	
		MIN V _{DD_IO} CLOAD = 60pf	TBD	–	TBD	V/ns	
PIXCLK frequency	^t PIXCLK	Default	30	–	60	MHz	3
PIXCLK to data valid	^t PD	Default	0.4 * ^t PIXCLK	0.5 * ^t PIXCLK	0.6 * ^t PIXCLK	ns	4
PIXCLK to FV HIGH	^t PFH	Default	0.4 * ^t PIXCLK	0.5 * ^t PIXCLK	0.6 * ^t PIXCLK	ns	4
PIXCLK to LV HIGH	^t PLH	Default	0.4 * ^t PIXCLK	0.5 * ^t PIXCLK	0.6 * ^t PIXCLK	ns	4
PIXCLK to FV LOW	^t PFL	Default	0.4 * ^t PIXCLK	0.5 * ^t PIXCLK	0.6 * ^t PIXCLK	ns	4
PIXCLK to LV LOW	^t PLL	Default	0.4 * ^t PIXCLK	0.5 * ^t PIXCLK	0.6 * ^t PIXCLK	ns	4
Input pin capacitance	C _{IN}		–	TBD	–	pF	

- Notes:
1. Jitter to be specified.
 2. PLL disabled.
 3. Should include information on adaptive pixel clock scheme if included.
 4. From falling edge of PIXCLK.



MT9M113: 1/6-Inch 1.3Mp SOC Digital Image Sensor Electrical Specifications

Table 42: DC Electrical Definitions and Characteristics—Parallel Mode

^fEXTCLK = 27 MHz; ^fPIXCLK = 60 MHz; V_{DD} = 1.8V; V_{DD_IO} = 1.8V and 2.8V; V_{AA} = 2.8V; V_{AA_PIX} = 2.8V; V_{DD_PLL} = 2.8V; V_{DD_PHY} = NA; C_{LOAD} = 30pF; T_A = 70°C; Lighting conditions = 90 Lux

Parameter	Symbol	Conditions	Min	Typ	Max	Units	Note
Core digital voltage	V _{DD}		1.7	1.8	1.95	V	
I/O digital voltage	V _{DD_IO1}		1.7	1.8	1.95	V	
I/O digital voltage	V _{DD_IO2}		2.5	2.8	3.1	V	
Analog voltage	V _{AA}		2.5	2.8	3.1	V	
Pixel supply voltage	V _{AA_PIX}		2.5	2.8	3.1	V	
PLL supply voltage	V _{DD_PLL}		2.5	2.8	3.1	V	
V _{DD} power supply rejection ratio	PSRR _{V_{DD}}	Frequency =	–	–	TBD	dB	
V _{AA} power supply rejection ratio	PSRR _{V_{AA}}	Frequency =	–	–	TBD	dB	
V _{AA_PIX} power supply rejection ratio	PSRR _{V_{AA_PIX}}	Frequency =	–	–	TBD	dB	
V _{DD_PLL} power supply rejection ratio	PSRR _{V_{DD_PLL}}	Frequency =	–	–	TBD	dB	
Digital operating current	I _{DD_1}	Context A/EXTCLK	–	14	–	mA	
I/O digital operating current (1.8V)	I _{DD_IO1}	Context A/EXTCLK	–	14	–	mA	
I/O digital operating current (2.8V)	I _{DD_IO2}	Context A/EXTCLK	–	25	–	mA	
Analog operating current	I _{AA_1}	Context A/EXTCLK	–	22	–	mA	
Pixel supply current	I _{AA_PIX1}	Context A/EXTCLK	–	1	–	mA	
PLL supply current	I _{DD_PLL1}	Context A/EXTCLK	–	NA	–	mA	
MIPI supply current	I _{DDIO_TX}	Context A/EXTCLK	–	NA	–	mA	
Total supply current		Context A/EXTCLK	–	37	–	mA	1
Total power consumption		Context A/EXTCLK	–	90	–	mW	1
Digital operating current	I _{DD_2}	Context B/EXTCLK	–	25	–	mA	
I/O digital operating current (1.8V)	I _{DD_IO1}	Context B/EXTCLK	–	9	–	mA	
I/O digital operating current (2.8V)	I _{DD_IO2}	Context B/EXTCLK	–	18	–	mA	
Analog operating current	I _{AA_2}	Context B/EXTCLK	–	35	–	mA	
Pixel supply current	I _{AA_PIX2}	Context B/EXTCLK	–	1	–	mA	
PLL supply current	I _{DD_PLL2}	Context B/EXTCLK	–	NA	–	mA	
MIPI supply current	I _{DDIO_TX}	Context B/EXTCLK	–	NA	–	mA	
Total supply current		Context B/EXTCLK	–	61	–	mW	1
Total power consumption		Context B/EXTCLK	–	146	–	mW	1
Digital operating current	I _{DD_1}	Context A/PLL	–	18	–	mA	
I/O digital operating current (1.8V)	I _{DD_IO1}	Context A/PLL	–	14	–	mA	
I/O digital operating current (2.8V)	I _{DD_IO2}	Context A/PLL	–	25	–	mA	
Analog operating current	I _{AA_1}	Context A/PLL	–	22	–	mA	
Pixel supply current	I _{AA_PIX1}	Context A/PLL	–	1	–	mA	
PLL supply current	I _{DD_PLL1}	Context A/PLL	–	21	–	mA	
MIPI supply current	I _{DDIO_TX}	Context A/PLL	–	NA	–	mA	
Total supply current		Context A/PLL	–	62	–	mA	1
Total power consumption		Context A/PLL	–	156	–	mW	1
Digital operating current	I _{DD_2}	Context B/PLL	–	28	–	mA	
I/O digital operating current (1.8V)	I _{DD_IO1}	Context B/PLL	–	9	–	mA	
I/O digital operating current (2.8V)	I _{DD_IO2}	Context B/PLL	–	18	–	mA	
Analog operating current	I _{AA_2}	Context B/PLL	–	36	–	mA	
Pixel supply current	I _{AA_PIX2}	Context B/PLL	–	1	–	mA	
PLL supply current	I _{DD_PLL2}	Context B/PLL	–	24	–	mA	



MT9M113: 1/6-Inch 1.3Mp SOC Digital Image Sensor Electrical Specifications

Table 42: DC Electrical Definitions and Characteristics—Parallel Mode (continued)

^fEXTCLK = 27 MHz; ^fPIXCLK = 60 MHz; V_{DD} = 1.8V; V_{DD_IO} = 1.8V and 2.8V; V_{AA} = 2.8V; V_{AA_PIX} = 2.8V; V_{DD_PLL} = 2.8V; V_{DD_PHY} = NA; C_{LOAD} = 30pF; T_A = 70°C; Lighting conditions = 90 Lux

Parameter	Symbol	Conditions	Min	Typ	Max	Units	Note
MIPI supply current	I _{DDIO_TX}	Context B/PLL	–	NA	–	mA	
Total supply current		Context B/PLL	–	89	–	mw	1
Total power consumption		Context B/PLL	–	222	–	mw	1
Total standby current							

Notes: 1. Total power consumption does not include current from V_{DD_IO}.

Table 43: DC Electrical Definitions and Characteristics—Serial Mode

^fEXTCLK = 27 MHz; ^fPIXCLK = 60 MHz; V_{DD} = 1.8V; V_{DD_IO} = 1.8V and 2.8V; V_{AA} = 2.8V; V_{AA_PIX} = 2.8V; V_{DD_PLL} = 2.8V; V_{DD_PHY} = NA; C_{LOAD} = 30pF; T_J = 25°C; Lighting conditions = 90 Lux

Parameter	Symbol	Conditions	Min	Typ	Max	Units	Note
Core digital voltage	V _{DD}		1.7	1.8	1.95	V	
I/O digital voltage	V _{DD_IO1}		1.7	1.8	1.95	V	
I/O digital voltage	V _{DD_IO2}		2.5	2.8	3.1	V	
Analog voltage	V _{AA}		2.5	2.8	3.1	V	
Pixel supply voltage	V _{AA_PIX}		2.5	2.8	3.1	V	
PLL supply voltage	V _{DD_PLL}		2.5	2.8	3.1	V	
V _{DD} power supply rejection ratio	PSRR _{VDD}	Frequency = TBD	–	–	TBD	dB	
V _{AA} power supply rejection ratio	PSRR _{VAA}	Frequency = TBD	–	–	TBD	dB	
V _{AA_PIX} power supply rejection ratio	PSRR _{VAA_PIX}	Frequency = TBD	–	–	TBD	dB	
V _{DD_PLL} power supply rejection ratio	PSRR _{VDD_PLL}	Frequency = TBD	–	–	TBD	dB	
Digital operating current	I _{DD_1}	Context A/EXTCLK	–	–	TBD	mA	
I/O digital operating current (1.8V)	I _{DD_IO1}	Context A/EXTCLK	–	–	TBD	mA	
I/O digital operating current (2.8V)	I _{DD_IO2}	Context A/EXTCLK	–	–	TBD	mA	
Analog operating current	I _{AA_1}	Context A/EXTCLK	–	–	TBD	mA	
Pixel supply current	I _{AA_PIX1}	Context A/EXTCLK	–	–	TBD	mA	
PLL supply current	I _{DD_PLL1}	Context A/EXTCLK	–	–	TBD	mA	
MIPI supply current	I _{DDIO_TX}	Context A/EXTCLK	–	–	TBD		
Total supply current		Context A/EXTCLK	–	–	TBD	mA	1
Total power consumption		Context A/EXTCLK	–	–	TBD	mW	1
Digital operating current	I _{DD_2}	Context B/EXTCLK	–	–	TBD	mA	
I/O digital operating current (1.8V)	I _{DD_IO1}	Context B/EXTCLK	–	–	TBD	mA	



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Table 43: DC Electrical Definitions and Characteristics—Serial Mode (continued)

^fEXTCLK = 27 MHz; ^fPIXCLK = 60 MHz; V_{DD} = 1.8V; V_{DD_IO} = 1.8V and 2.8V; V_{AA} = 2.8V; V_{AA_PIX} = 2.8V; V_{DD_PLL} = 2.8V; V_{DD_PHY} = NA; C_{LOAD} = 30pF; T_J = 25°C; Lighting conditions = 90 Lux

Parameter	Symbol	Conditions	Min	Typ	Max	Units	Note
I/O digital operating current (2.8V)	I _{DD_IO2}	Context B/EXTCLK	–	–	TBD	mA	
Analog operating current	I _{AA_2}	Context B/EXTCLK	–	–	TBD	mA	
Pixel supply current	I _{AA_PIX2}	Context B/EXTCLK	–	–	TBD	mA	
PLL supply current	I _{DD_PLL2}	Context B/EXTCLK	–	–	TBD	mA	
MIPI supply current	I _{DDIO_TX}	Context B/EXTCLK	–	–	TBD	mA	
Total supply current		Context B/EXTCLK	–	–	TBD	mW	1
Total power consumption		Context B/EXTCLK	–	–	TBD	mW	1
Digital operating current	I _{DD_1}	Context A/PLL	–	–	TBD	mA	
I/O digital operating current (1.8V)	I _{DD_IO1}	Context A/PLL	–	–	TBD	mA	
I/O digital operating current (2.8V)	I _{DD_IO2}	Context A/PLL	–	–	TBD	mA	
Analog operating current	I _{AA_1}	Context A/PLL	–	–	TBD	mA	
Pixel supply current	I _{AA_PIX1}	Context A/PLL	–	–	TBD	mA	
PLL supply current	I _{DD_PLL1}	Context A/PLL	–	–	TBD	mA	
MIPI supply current	I _{DDIO_TX}	Context A/PLL	–	–	TBD	mA	
Total supply current		Context A/PLL	–	–	TBD	mA	1
Total power consumption		Context A/PLL	–	–	TBD	mW	1
Digital operating current	I _{DD_2}	Context B/PLL	–	–	TBD	mA	
I/O digital operating current (1.8V)	I _{DD_IO1}	Context B/PLL	–	–	TBD	mA	
I/O digital operating current (2.8V)	I _{DD_IO2}	Context B/PLL	–	–	TBD	mA	
Analog operating current	I _{AA_2}	Context B/PLL	–	–	TBD	mA	
Pixel supply current	I _{AA_PIX2}	Context B/PLL	–	–	TBD	mA	
PLL supply current	I _{DD_PLL2}	Context B/PLL	–	–	TBD	mA	
MIPI supply current	I _{DDIO_TX}	Context B/PLL	–	–	TBD	mA	
Total supply current		Context B/PLL	–	–	TBD	mW	1
Total power consumption		Context B/PLL	–	–	TBD	mW	1
Maximum total standby current	I _{TOTAL_STBY}	Context A/B	–	–	20	μA	

Notes: 1. Total power consumption does not include current from V_{DD_IO}.



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Table 44: I/O Parameters

Parameter	Symbol	Conditions	Min	Typ	Max	Units
Input HIGH voltage	V _{IH}	At specified I _{IN}	V _{DD_IO} – 0.4		V _{DD_IO} + 0.3	V
Input LOW voltage	V _{IL}	V _{DD_IO} = 2.8V at specified I _{IN}	–0.3		0.6	V
		V _{DD_IO} = 1.8V at specified I _{IN}	–0.3		0.4	
Input leakage current	I _{IN}	No pull-up resistor; V _{IN} = V _{DD} or DGND	–50		50	μA
Output HIGH voltage	V _{OH}	At specified I _{OH}	V _{DD_IO} – 0.4		–	V
Output LOW voltage	V _{OL}	At specified I _{OL}	–		0.4	V
Output HIGH current	I _{OH}	At minimum of 1.4V V _{OH}	–		–13	mA
		At minimum of 2.4V V _{OH}	–		–20	mA
Output LOW current	I _{OL}	At maximum of V _{OL}	–		12	mA
		At specified V _{OL}	–		15	mA
Tri-state output leakage current	I _{OZ}				5	μA

Table 45: Power Consumptions at Different Resolutions

f_{EXTCLK} = 27 MHz; f_{PIXCLK} = 60 MHz; V_{DD} = 1.8V; V_{DD_IO} = 1.8V and 2.8V; V_{AA} = 2.8V; V_{AA_PIX} = 2.8V; V_{DD_PLL} = 2.8V; V_{DD_PHY} = NA; C_{LOAD} = 30pF; T_J = 25°C; Lighting conditions = 90 Lux

Load (pF)	Temp (°C)	Frame Rate	Min	Typ	Max	Units	Note
15	Minimum						
30	Minimum						
68	Minimum						
15	Ambient						
30	Ambient						
68	Ambient						
15	Maximum						
30	Maximum						
68	Maximum						



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Caution

Note: Stresses above those listed may cause permanent damage to the device. This is a stress rating only, and functional operation of the device at these or any other conditions above those indicated in the product specification is not implied. Exposure to absolute maximum rating conditions for extended periods may affect device reliability.

Table 46: Absolute Maximum Ratings

Symbol	Parameter	Rating		Units
		Min	Max	
V _{DD_MAX}	Core digital voltage	-0.3	2.4	V
V _{DD_IO_MAX}	I/O digital voltage	-0.3	4.0	V
V _{AA_MAX}	Analog voltage	-0.3	4.0	V
V _{AA_PIX_MAX}	Pixel supply voltage	-0.3	4.0	V
V _{DD_PLL_MAX}	PLL supply voltage	-0.3	4.0	V
V _{IH_MAX}	Input HIGH voltage	-0.3	V _{DD_IO} + 0.3	V
V _{IL_MAX}	Input LOW voltage	-0.3	–	V
T _{OP}	Operating temperature (measure at junction)	-30	75	°C
T _{STG}	Storage temperature	-40	85	°C



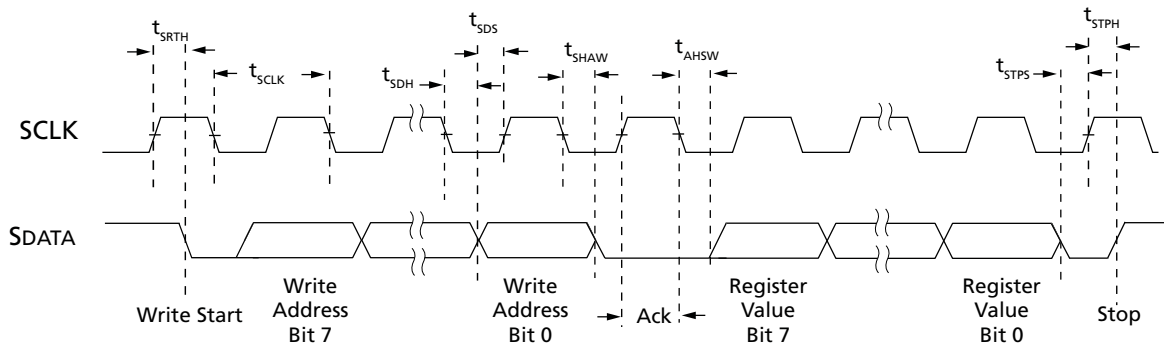
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Two-Wire Serial Bus Timing

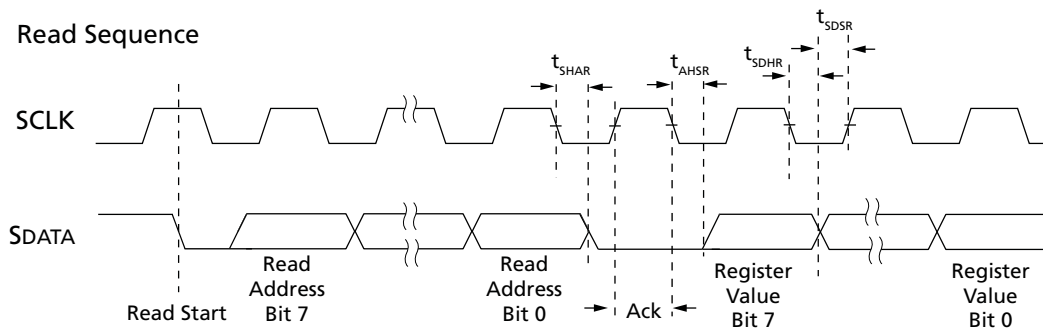
Figure 37 and Table 47 on page 119 describe the timing for the two-wire serial interface.

Figure 37: Two-Wire Serial Bus Timing Parameters

Write Sequence



Read Sequence





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Table 47: Two-Wire Serial Bus Characteristics

^fEXTCLK = 8–54 MHz; V_{DD} = 1.8V; V_{DD_IO} = 1.8 and 2.8V; V_{AA} = 2.8V; V_{AA_PIX} = 2.8V;
V_{DD_PLL} = 2.8V; T_J = 25°C; C_{LOAD} = 30pF

Parameter	Symbol	Conditions	Min	Typ	Max	Units	Notes
Serial interface input clock frequency	^f SCLK		100	–	400	kHz	
Serial interface input clock period	^t SCLK		2.5	–	10	μs	Master clock cycle units or PLL cycles if enabled
SCLK duty cycle			40	50	60	%	
Start hold time	^t SRTTH	Write/Read	0.6	–	–	μs	Master clock cycle units or PLL cycles if enabled
S _{DATA} hold	^t SDH	Write	0.6	–	–	μs	Master clock cycle units or PLL cycles if enabled
S _{DATA} setup	^t SDS	Write	0.6	–	–	μs	Master clock cycle units or PLL cycles if enabled
S _{DATA} hold to ack	^t SHAW	Write	0.6	–	–	μs	Master clock cycle units or PLL cycles if enabled
Ack hold to S _{DATA}	^t AHSW	Write	0.6	–	–	μs	Master clock cycle units or PLL cycles if enabled
Stop setup time	^t STPS	Write/Read	0.6	–	–	μs	Master clock cycle units or PLL cycles if enabled
Stop hold time	^t STPH	Write/Read	0.6	–	–	μs	Master clock cycle units or PLL cycles if enabled
S _{DATA} hold to ack	^t SHAR	Read	0.3	–	1	μs	Master clock cycle units or PLL cycles if enabled
Ack hold to S _{DATA}	^t AHSR	Read	0.3	–	1	μs	Master clock cycle units or PLL cycles if enabled
S _{DATA} hold	^t SDHR	Read	0.3	–	1	μs	Master clock cycle units or PLL cycles if enabled
S _{DATA} setup	^t SDSR	Read	0.1	–	1	μs	Master clock cycle units or PLL cycles if enabled
No programmable skew	S _{DATA} Slew Rate	MAX V _{DD_IO} C _{LOAD} = 15pF	–	–	–	V/ns	
		TYP V _{DD_IO} C _{LOAD} = 30pF	–	–	–	V/ns	
		MIN V _{DD_IO} C _{LOAD} = 60pF	–	–	–	V/ns	
Programmable skew Condition 1		MAX V _{DD_IO} C _{LOAD} = 15pF	–	–	–	V/ns	
		TYP V _{DD_IO} C _{LOAD} = 30pF	–	–	–	V/ns	
		MIN V _{DD_IO} C _{LOAD} = 60pF	–	–	–	V/ns	
Programmable skew Condition 2		MAX V _{DD_IO} C _{LOAD} = 15pF	–	–	–	V/ns	
		TYP V _{DD_IO} C _{LOAD} = 30pF	–	–	–	V/ns	
		MIN V _{DD_IO} C _{LOAD} = 60pF	–	–	–	V/ns	
Serial interface input pin capacitance	C _{IN_SI}		–	–	–	pF	



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Table 47: Two-Wire Serial Bus Characteristics (continued)

^fEXTCLK = 8–54 MHz; V_{DD} = 1.8V; V_{DD_IO} = 1.8 and 2.8V; V_{AA} = 2.8V; V_{AA_PIX} = 2.8V;
V_{DD_PLL} = 2.8V; T_J = 25°C; C_{LOAD} = 30pF

Parameter	Symbol	Conditions	Min	Typ	Max	Units	Notes
SDATA max load capacitance	C _{LOAD_SD}		–	–	30	pF	
SDATA pull-up resistor	R _{SD}		–	1.5	–	KΩ	



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Advance: This data sheet contains initial descriptions of products still under development.



MT9M113: 1/6-Inch 1.3Mp SOC Digital Image Sensor Revision History

Revision History

Rev. B, Advance	3/13/2007
- Updated Table 10 on page 47 through Table 33 on page 101 with new data from RegDB. Edited RegDB format. - EXTCLK minimum changed from 6 MHz to 8 MHz in various places. - Updated electrical timing from Figure 29 on page 103 through Figure 37 on page 118. - Updated electrical data from Table 34 on page 103 through Table 47 on page 119. - Added DigitalClarity to trademarks on last page.	
Rev. A, Advance	10/06
Initial release	